

NTL-LTER Lake Datasets

Summary

This dataset was prepared for Environmental Data Analytics (ENV 872L) at Duke University, Spring 2020

The dataset contains data from studies on several lakes in the North Temperate Lakes District in Wisconsin, USA. Data were collected as part of the Long Term Ecological Research station established by the National Science Foundation.

Database Information

Data were collected from the North Temperate Lakes Long Term Ecological Research website. More information can be found here: <https://lter.limnology.wisc.edu/history/>

Data were collected using the Data tool (<https://lter.limnology.wisc.edu/regional-lakes-survey/>).

From the Data homepage, the following selections were made:

- Cascade (NTL Categories)
- [North Temperate Lakes LTER Regional Survey Water Chemistry 2015 – current](#)

AND

- Cascade Project at North Temperate Lakes LTER Core Data Nutrients 1991 - 2016

AND

- Cascade Project at North Temperate Lakes LTER Core Data Physical and Chemical Limnology 1984 - 2016
- On each of the three pages, data were downloaded via the EDI Data Portal

csv files were saved as `NTL-LTER_Lake_Carbon_Raw.csv`, `NTL-LTER_Lake_ChemistryPhysics_Raw.csv`, and `NTL-LTER_Lake_Nutrients_Raw.csv`.

Data were accessed 2018-12-06.

Data Content Information

From the NTL-LTER site:

Zooplankton

Carbon

Data on dissolved organic and inorganic carbon, particulate organic matter, partial pressure of CO₂ and absorbance at 440nm. Samples were collected with a Van Dorn sampler. Organic carbon and absorbance samples were collected from the epilimnion, metalimnion, and hypolimnion. Inorganic samples were collected at depths corresponding to 100%, 50%, 25%, 10%, 5%, and 1% of surface irradiance, as well as one sample from the hypolimnion. Samples for the partial pressure of CO₂ were collected from two meters above the lake surface (air) and just below the lake surface (water). Sampling frequency: varies; number of sites: 14

Detailed field and laboratory protocols can be found in the Cascade Methods Manual, found here: https://cascade.limnology.wisc.edu/public/public_files/methods/CascadeMa...

POC, PON and DOC: 1. 100 - 300 ml (Typically ~200mL for PML, 150 metalimnion and 75 – 100 for the hypolimnion) of lake water from each depth was filtered through 153 um mesh to remove large zooplankton. Water was then filtered through a precombusted 25mm GF/F filter (0.7 um pore size) at less than 200 mm Hg pressure. Filters were placed in drying oven at 60 C to dry for at least 48 hours. 20mL of filtered water was stored in a scintillation vial and acidified with 200uL of 2N H₂SO₄ for DOC analysis. Blank samples for POC and DOC were prepared with deionized water to control for contamination. All samples were sent to the Cary Institute of Ecosystem Studies for analysis.

Absorbance: 60ml of water was filtered through a 25mm GF/F filter and refrigerated until it was able to be run. Samples were warmed up to room temperature and run on a spectrophotometer in a 10cm glass cuvette. The spectrophotometer was set to 440nm and blanked with deionized water. The cuvette was rinsed once with sample water and then filled and absorbance was measured.

DIC: Water was sampled with a van dorn and taken back to the lab. 10 mL subsamples were taken with syringes and 200 uL of 2N H₂SO₄ and 20 mL of helium gas were added to the syringe. Syringes were shaken for one minute and 10 mL of the helium headspace was injected into a Gas Chromatograph with Thermal Conductivity Detector to determine DIC concentration.

PCO₂: Air PCO₂ was measured by filling a syringe with air from two meters above the lake surface and running though the GC. Water PCO₂ was measured by filling a two liter bottle with lake surface water and replacing 60 mL of water with atmospheric air for headspace. The bottle was shaken for 100 seconds and two subsamples of the headspace were taken and run though a GC in the lab.

Nutrients

Physical and chemical variables are measured at one central station near the deepest point of each lake. In most cases these measurements are made in the morning (0800 to 0900). Vertical profiles are taken at varied depth intervals. Chemical measurements are sometimes made in a pooled mixed layer sample (PML); sometimes in the epilimnion, metalimnion, and hypolimnion; and sometimes in vertical profiles. In the latter case, depths for sampling usually correspond to the surface plus depths of 50percent, 25percent, 10percent, 5percent and 1percent of surface irradiance. The 1991-1999 chemistry data was obtained from the Lachat auto-analyzer. Like the process data, there are up to seven samples per sampling date due to Van Dorn collections across a depth interval according to percent irradiance. Voichick and LeBouton (1994) describe the autoanalyzer procedures in detail. Nutrient samples were sent to the Cary Institute of Ecosystem Studies for analysis beginning in 2000. The Kjeldahl method for measuring nitrogen is not used at IES, and so measurements reported from 2000 onwards are Total Nitrogen.

Methods for 1984-1990 were described by Carpenter and Kitchell (1993) and methods for 1991-1997 were described by Carpenter et al. (2001).

Carpenter, S.R. and J.F. Kitchell (eds.). 1993. *The Trophic Cascade in Lakes*. Cambridge University Press, Cambridge, England.

Carpenter, S.R., J.J. Cole, J.R. Hodgson, J.F. Kitchell, M.L. Pace, D. Bade, K.L. Cottingham, T.E. Essington, J.N. Houser and D.E. Schindler. 2001. Trophic cascades, nutrients and lake productivity: whole-lake experiments. *Ecological Monographs* 71: 163-186.

Units

Variable	Description	Typical Units	Notes
TP	Total Phosphorus	µg P/L	Includes dissolved + particulate P; measured after digestion.
SRP (PO ₄ ³⁻ -P)	Soluble Reactive Phosphorus	µg P/L	Also called "orthophosphate."
TN	Total Nitrogen	mg N/L	Sum of organic + inorganic N; may be converted from Kjeldahl + NO ₃ ⁻ /NO ₂ ⁻ .
NO ₃ ⁻ -N	Nitrate Nitrogen	mg N/L	Sometimes reported in µg/L; check dataset metadata.
NH ₄ ⁺ -N	Ammonium Nitrogen	mg N/L	Filtered sample, colorimetric or autoanalyzer method.
SiO ₂	Dissolved Silica	mg Si/L or µmol Si/L	Used in productivity and diatom studies.
DOC	Dissolved Organic Carbon	mg C/L	Often accompanies nutrient datasets.

Physical and chemical limnology

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Naming conventions and file formats

Files are named according to the following naming convention: `database_name_datatype_details_stage.format`, where:

database_name refers to the database from where the data originated

datatype is a description of data

details are additional descriptive details, particularly important for processed data

stage refers to the stage in data management pipelines (e.g., raw, cleaned, or processed)

format is a non-proprietary file format (e.g., .csv, .txt)

Additional Information and Support

For more information, please contact the data assembler, **Kateri Salk** (kateri.salk@duke.edu)